Multivariate Calculus Lecture Notes
Section 12.5 – Planes in Three-Space

I. Defining a plane, geometrically
   a. Three points will work.
   b. Two intersecting but non-parallel lines will work.
   c. A point and a vector perpendicular to the plane will work. (Illustrate this.)
   d. A point and a vector in the plane are not sufficient. (Illustrate this, too.)

II. Derivation of plane equation
   a. Let (x_0, y_0, z_0) be a fixed point in a plane.
   b. Let $\mathbf{n} = \langle A, B, C \rangle$ be a vector perpendicular to the plane.
   c. What property characterizes whether the point (x, y, z) is in the plane?
      i. The vector between (x, y, z) and (x_0, y_0, z_0) must be in the plane.
      ii. If so, this vector must be orthogonal to $\mathbf{n}$.
      iii. Thus we must have $\mathbf{n} \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$.
   d. The standard equation of a plane in space is therefore given by $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$, where $\langle A, B, C \rangle$ is a vector normal to the plane and (x_0, y_0, z_0) is a point in the plane.
   e. Your book also mentions a vector form of the plane equation.
      i. Take $\mathbf{n} \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$, and multiply through to get $\mathbf{n} \cdot \langle x, y, z \rangle = \mathbf{n} \cdot \langle x_0, y_0, z_0 \rangle$.
      ii. The *vector form* of the plane equation is then $\mathbf{n} \cdot \langle x, y, z \rangle = d$, where $d = \mathbf{n} \cdot \langle x_0, y_0, z_0 \rangle$.

III. Examples
   a. How do you tell if two planes are parallel?
      - They have the same normal vectors.
   b. Find an equation describing the plane through the points $(1, 0, 2)$, $(3, -1, 6)$, and $(5, 2, 4)$.
      i. First we need a point in the plane; $(1, 0, 2)$ will do nicely.
      ii. Second, we need a vector normal to the plane.
         - If we form two vectors in the plane from the points we have and take their cross product then we'll have a vector normal to the plane.
         - Two vectors in the plane: $\langle 2, -1, 4 \rangle$ and $\langle 4, 2, 2 \rangle$
         - Cross product:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 4 \\ 4 & 2 & 2 \end{vmatrix} = (-2 - 8)\mathbf{i} - (4 - 16)\mathbf{j} + (4 + 4)\mathbf{k} = -10\mathbf{i} + 12\mathbf{j} + 8\mathbf{k}.$$

      iii. Thus the plane can be expressed as $-10(x - 1) + 12y + 8(z - 2) = 0$.
      iv. Alternatively, this can be written as $-10x + 12y + 8z - 6 = 0$.
      v. Crossing the vectors the other way yields $10x - 12y - 8z + 6 = 0$.
   c. Find the intersection of the line given by $x = 1 - t$, $y = 3t$, $z = 1 + t$ and the plane given by $2x - y + 3z = 6$.
      i. In order for a line to intersect a plane there has to be some value of t that yields a point on the line that is in the plane.
      ii. In other words, there must be some value of t that produces a point that satisfies the plane equation.
      iii. Substitute: $2(1 - t) - 3t + 3(1 + t) = 6 \implies 2 - 2t - 3t + 3 + 3t = 6 \implies -2t = 1 \implies t = -1/2$.

iv. This yields the point $(3/2, -3/2, 1/2)$.
v. Is this point in the plane?
1. Does $2(3/2) + 3/2 + 3/2 = 6$?
2. Yes.

d. Find the line of intersection of the planes $x + y + z = 1$ and $x - 2y + 3z = 1$.
i. First, realize that we've only got two equations and so we're going to have a free parameter.
1. Moreover, we have some choice in that parameter.
2. Different choices for the parameter will start the line in different places or trace the line out faster or slower, but they won't change the line.
ii. Solve for x in both equations: $x = 1 - y - z$ and $x = 1 + 2y - 3z$.
iii. Set equal and solve for y: $1 - y - z = 1 + 2y - 3z \implies 3y = 2z \implies y = \frac{2}{3}z$.
iv. If we express x in terms of z we obtain $x = 1 - \frac{2}{3}z - z = 1 - \frac{5}{3}z$.
v. The easiest thing to do at this point is to let $z = t$.
vi. Then we have $x = 1 - \frac{5}{3}t$, $y = \frac{2}{3}t$, $z = t$ for our parametric equations.

Homework: Section 12.5 #1, 4, 10, 11, 15, 20, 24, 27, 40, 57, 63.